

A Data-Driven Analysis of Customer Churn Risk in Retail Banking

1. Background and Problem Statement

Customer churn is a critical issue in the retail banking industry because it directly affects profitability, customer lifetime value, and long-term growth. Retaining existing customers is generally far more cost-effective than acquiring new ones, making churn reduction a strategic priority for banks. In an increasingly competitive and digital-first environment, customers can easily switch to alternative financial institutions that offer better service quality, more convenient digital platforms, or more attractive financial products.

In this project, the business context focuses on a retail bank facing rising customer attrition in its personal banking segment. The bank's traditional retention approaches, such as mass email campaigns and broad outreach efforts, are no longer sufficient in addressing churn effectively. At the same time, limited use of analytics makes it difficult for the bank to identify at-risk customers early and intervene in a targeted way.

The objective of this project is therefore to build a predictive analytics solution that can identify customers who are likely to churn and uncover the factors that drive churn behaviour. These insights can then support more focused retention strategies, better resource allocation, and improved customer relationship management. Rather than relying on generic engagement efforts, the bank can use model-driven insights to take earlier and more personalised action.

2. Dataset Overview

The analysis was conducted using a cleaned bank customer dataset containing 10,000 observations and 15 variables. The target variable is *Exited*, where a value of 1 indicates that a customer churned and 0 indicates that the customer remained with the bank. Predictor variables include customer demographics, relationship characteristics, financial standing, and engagement indicators such as age, geography, gender, number of products, account balance, credit score, tenure, activity status, and card type.

Initial data inspection showed that the dataset did not contain missing values or duplicate records. Variables such as customer ID and surname were excluded because they served only as identifiers and did not provide predictive value. One important issue discovered during exploration was that the variable *Complain* was almost perfectly correlated with the churn outcome. This indicated serious data leakage risk, so it was removed before model development to avoid unrealistically inflated model performance.

The churn rate in the dataset is approximately 20.4%, meaning the data is imbalanced, with substantially more non-churned than churned customers. Because of this, the project applied SMOTE on the training set only to improve class balance during model training while preserving a fair evaluation on the test set.

3. Exploratory Data Analysis

The exploratory analysis revealed several meaningful churn patterns.

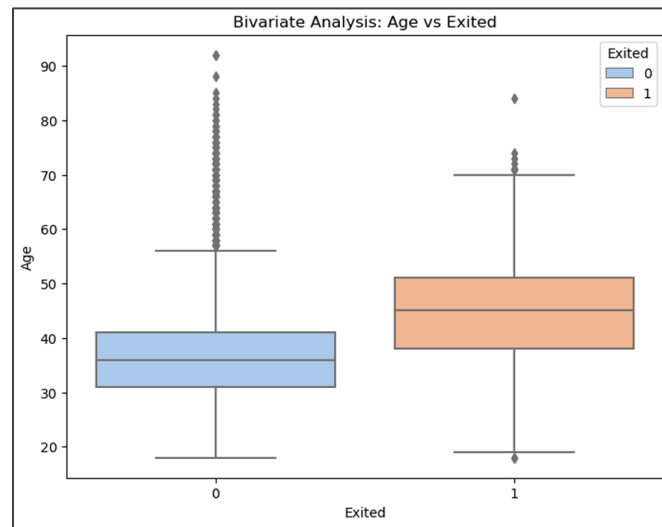


Figure 1: Boxplots of “Age” vs “Exited”

First, age appears to be a strong factor. Customers who churned were generally older than those who stayed. The median age of churned customers was around 45, compared with 36 for retained customers. This suggests that older customers may be more likely to leave, possibly due to changing financial needs or dissatisfaction with the bank’s offerings.

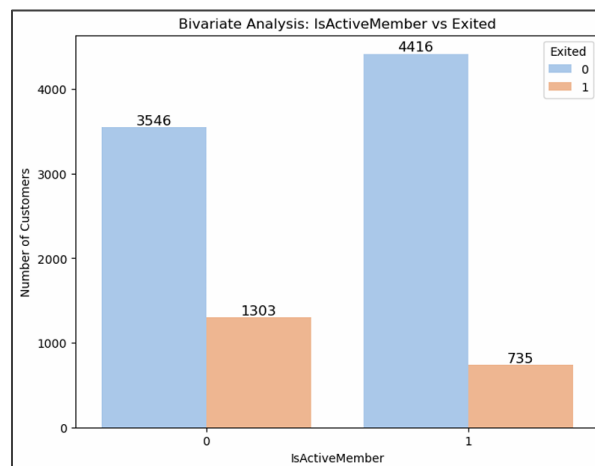


Figure 2: Barchart of “IsActiveMember” vs “Exited”

Second, customer engagement plays an important role. Inactive customers had a much higher churn rate than active customers. The churn rate for inactive customers was about 26.9%, compared with 14.3% for active customers. This indicates that activity level is a strong signal of retention and that customers who are less engaged with the bank are significantly more likely to exit.

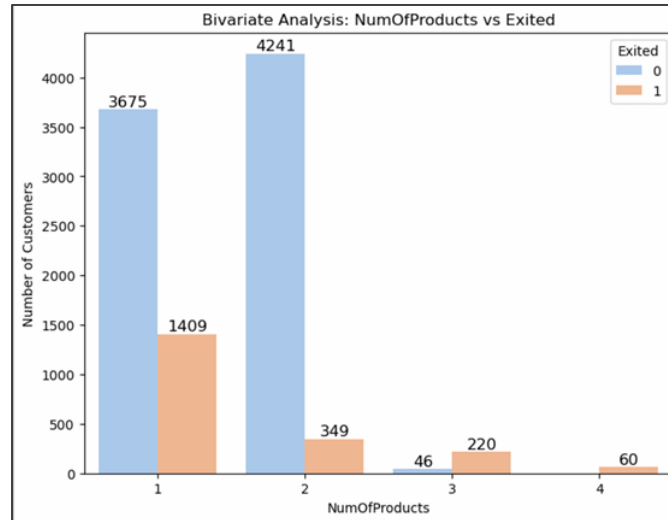


Figure 3: Barchart of “NumOfProducts” vs “Exited”

Third, number of products showed an interesting relationship with churn. Customers with two products had the lowest churn rate, while those with three or four products showed extremely high churn rates. This may suggest that cross-selling is beneficial only up to a certain point, after which customers may perceive the product mix as overly complex or less valuable.

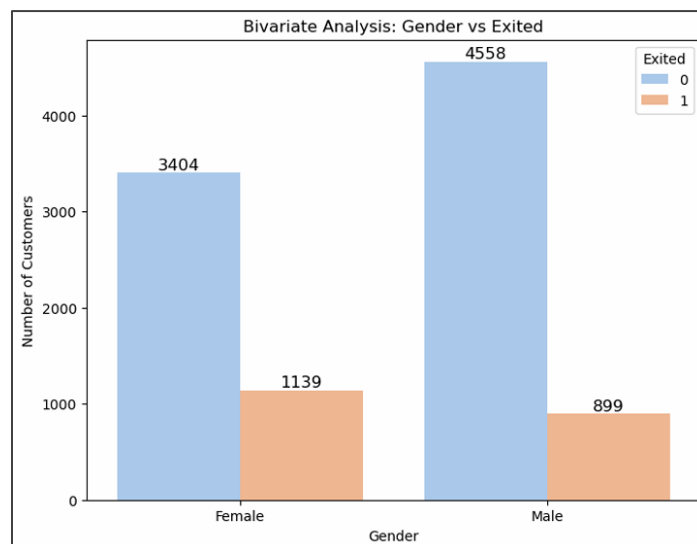


Figure 4: Barchart of “Gender” vs “Exited”

Fourth, gender also showed some variation, with female customers exhibiting a higher churn rate than male customers in the dataset. While this does not imply causation, it indicates that gender-related differences in customer experience or financial behaviour may warrant further investigation.

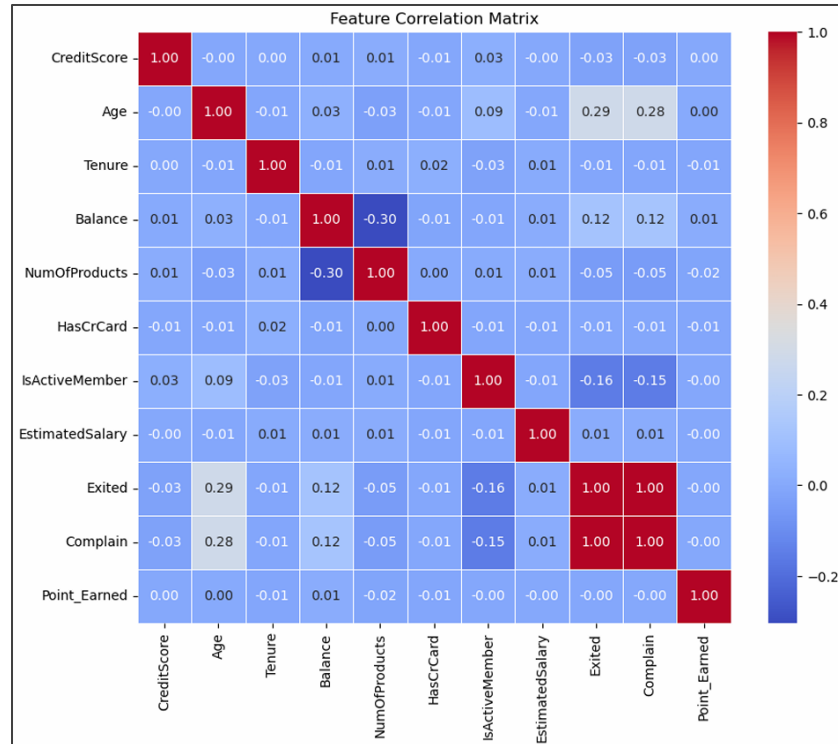


Figure 4: Feature Correlation Matrix

Finally, the correlation analysis showed that Age and Balance were positively associated with churn, while IsActiveMember was negatively associated with churn. The strongest warning sign from EDA was the near-perfect correlation between Complain and Exited, reinforcing the need to remove that feature before modelling.

Overall, the EDA suggests that churn is not random. It is associated with a combination of demographic, behavioural, and relationship-related factors, which makes machine learning models appropriate for this prediction task.

4. Analytical Approach

To predict customer churn, three classification models were developed and compared: **Logistic Regression**, **Random Forest**, and **XGBoost**. These models were selected to represent a range of analytical complexity, from interpretable statistical modelling to more advanced ensemble methods.

Before modelling, the dataset was split into training and testing sets using a 70:30 ratio. Since the target variable was imbalanced, SMOTE was applied only to the training data to generate a more balanced class distribution for model learning. This helps reduce bias toward the majority class while ensuring the test set remains representative of real-world data. Categorical variables were encoded where required, and the leaked variable Complain was removed prior to training.

The three models were evaluated using classification metrics including accuracy, AUC, precision, recall, and F1 score, depending on the model output available. This allowed for a more complete comparison beyond simple accuracy alone. In churn prediction, accuracy is useful, but recall and AUC are especially important because the bank needs to identify churners effectively without overlooking too many at-risk customers.

5. Model Performance

Model Performance Comparison:						
	Model	Accuracy	Precision	Recall	F1	AUC
2	XGBoost (Tuned)	0.872000	0.763341	0.538462	0.631478	0.873077
1	Random Forest (Tuned)	0.869667	0.769608	0.513912	0.616290	0.861417
0	Logistic Regression	0.698667	0.369546	0.679214	0.478662	0.757989

Figure 5: Model Performance

Among the three models, XGBoost performed best overall, followed closely by Random Forest, while Logistic Regression delivered the weakest predictive performance.

The Logistic Regression model achieved an accuracy of 69.87% and an AUC of approximately 0.758. This suggests that the model had fair predictive power, but it was less effective than the ensemble methods at separating churners from non-churners. Its main strength lies in interpretability rather than predictive accuracy.

The Random Forest model achieved an accuracy of 86.97%, F1 score of 0.616 and an AUC of 0.861. These results indicate that Random Forest was highly effective in identifying churners while maintaining a strong balance between false positives and false negatives.

The XGBoost model achieved the best results, with an accuracy of 87.2%, an F1 score of 0.631, and an AUC of 0.873. This makes it the strongest model in terms of overall discrimination and classification performance. Compared with Random Forest, XGBoost delivered slightly better predictive accuracy and AUC, suggesting that it captured complex churn patterns more effectively.

In summary, Logistic Regression is useful as a baseline and for interpretation, but the ensemble models clearly outperform it. Between the two advanced models, XGBoost is the recommended final model because it provides the most balanced and reliable performance for churn prediction.

6. Key Drivers of Bank Churn Risk

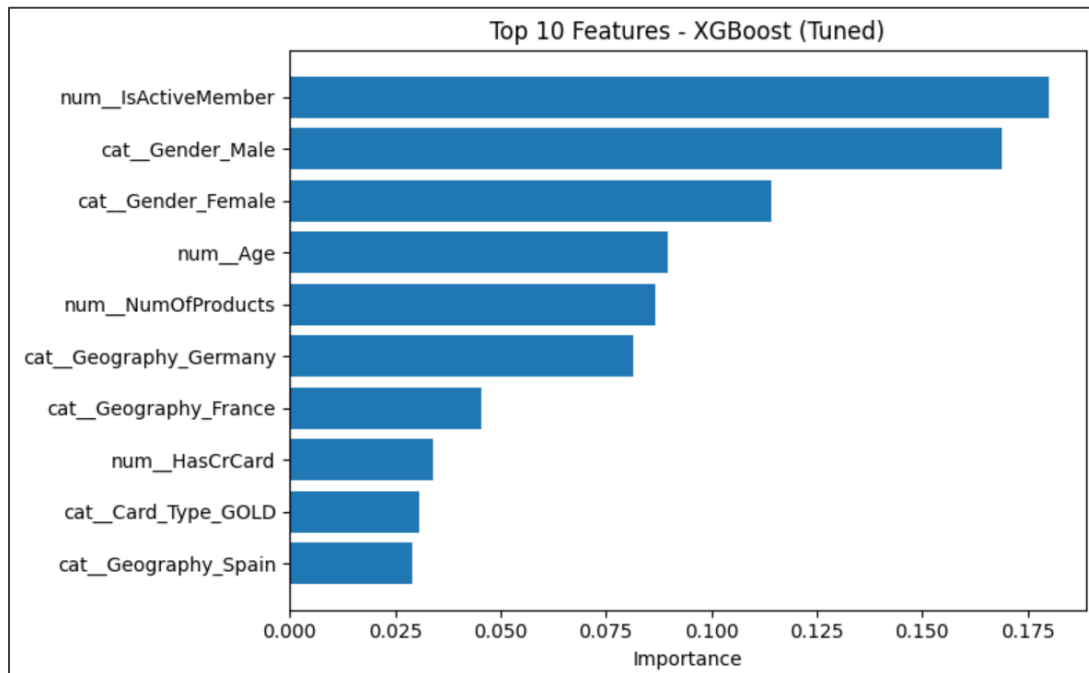


Figure 6: Feature Importances Analysis from tuned XGBoost model

The feature importance analysis from the tuned XGBoost model shows that customer churn is mainly driven by a combination of engagement, demographic factors, relationship depth, and geography. The most important variable is **IsActiveMember**, followed by gender, age, number of products, and geography. Other relevant but less influential factors include credit card ownership and gold card type.

Among all features, **IsActiveMember** is the strongest predictor, suggesting that customer engagement is the most critical factor in churn. Customers who are less active are much more likely to leave the bank, making inactivity an important warning sign. This highlights that churn is not only influenced by who the customer is, but also by how actively they use the bank's services.

Gender also appears as an important predictor, with both male and female dummy variables ranked highly in the model. This suggests that churn behaviour differs across customer groups and that different segments may respond differently to products, services, or communication strategies.

Age is another major driver, indicating that life stage and financial needs influence churn risk. In addition, **NumOfProducts** is important, showing that the number of products a customer holds affects retention. While more products may indicate a deeper relationship, they do not always guarantee loyalty. The model also highlights geography, especially Germany, as an important factor, suggesting that churn patterns vary by market. Finally, **HasCrCard** and **Card Type_GOLD** have some influence, though they are less important than the main drivers.

7. Business Implications

The XGBoost feature importance results provide several practical implications for the bank's retention strategy.

First, since `IsActiveMember` is the strongest predictor of churn, the bank should focus on improving customer engagement. Less active customers should be identified early and targeted with re-engagement efforts such as personalised reminders, digital prompts, loyalty rewards, and proactive service follow-ups. This suggests that increasing activity levels could have the greatest impact on reducing churn.

Second, the importance of gender indicates that retention strategies may need to be more segmented. The bank should not assume that all customers respond in the same way to products, services, or communication. Instead, it should test targeted approaches across different customer groups to better match their needs and preferences.

Third, age is also a strong driver, which means retention efforts should consider life stage differences. Younger customers may value convenience, flexibility, and digital features, while older customers may place more importance on trust, relationship quality, and personalised support. A single retention approach is therefore unlikely to work for all customers.

Fourth, the significance of `NumOfProducts` suggests that cross-selling should be handled carefully. Although holding more products can strengthen customer relationships, it does not always lead to higher loyalty. The bank should focus on offering relevant and valuable products rather than simply increasing product count.

Fifth, the role of geography, especially Germany, suggests that churn patterns differ across markets. Regional differences may reflect local competition, customer expectations, or service quality, so the bank should adapt retention strategies to country-level conditions instead of relying fully on a centralised approach.

Lastly, credit card ownership and gold card type also contribute to churn prediction, though to a lesser extent. These features may help the bank identify premium or card-related customer segments with different expectations and risks.

Overall, the bank should adopt a more targeted, data-driven strategy. The strongest focus should be on inactive customers, while also taking into account segmentation by age, gender, product profile, and geography to make retention efforts more precise and effective.

8. Conclusion

This project shows that machine learning can support more effective churn management in retail banking. Among the models tested, XGBoost performed best, offering the strongest predictive performance and the clearest insight into the key drivers of churn.

The results show that churn is mainly influenced by customer activity, gender, age, number of products, and geography. In particular, inactive customers represent the highest-risk group, making engagement the most important area for intervention.

Overall, the project provides both predictive and business value. By using XGBoost to identify at-risk customers and acting on the main churn drivers, the bank can improve retention efforts, reduce customer loss, and support stronger long-term profitability.